DO NOW — RECALL

# Lesson 121 · Writing Algorithms with Validation

Name: _____ Class: _____ Date: _____

**1. Why must a trace table record changes rather than only final output?**

**2. After pushing A then B onto a stack, which is popped first?**

**3. Why must a binary search operate on ordered data?**

**4. An OR gate has inputs 0 and 1. What is the output?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

It reveals execution path and when variable values change.

### 2. Model answer

B, because a stack is LIFO.

### 3. Model answer

It discards half the search space using an ordering comparison.

### 4. Model answer

1.

### By the end, you can…

• State valid range, type and required behaviour before coding.

• Use sequence, selection and iteration with clear termination.

• Use normal, boundary, erroneous and extreme cases appropriately.

### Vocabulary — know these terms

validation — Checking that input meets rules.

robustness — Ability to handle invalid/unexpected input safely.

iteration — Repetition based on loop condition.

boundary — Input at/near permitted limit.

NEW LEARNING

# Read, annotate and explain

### Define constraints

State valid range, type and required behaviour before coding.

### Build control flow

Use sequence, selection and iteration with clear termination.

### Prove with tests

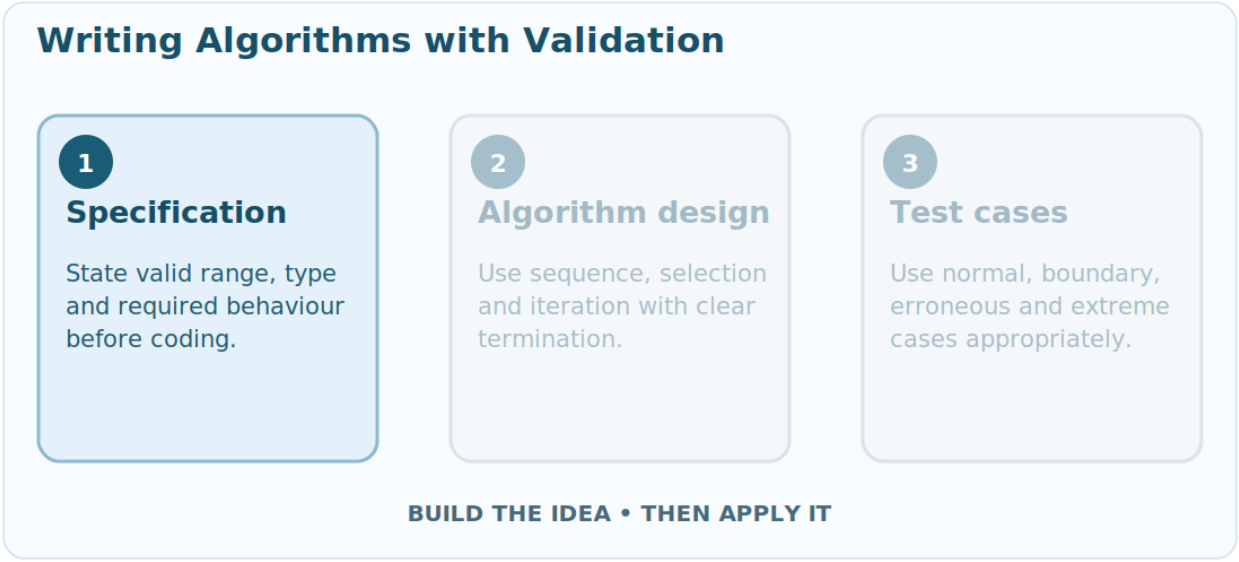
Use normal, boundary, erroneous and extreme cases appropriately.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

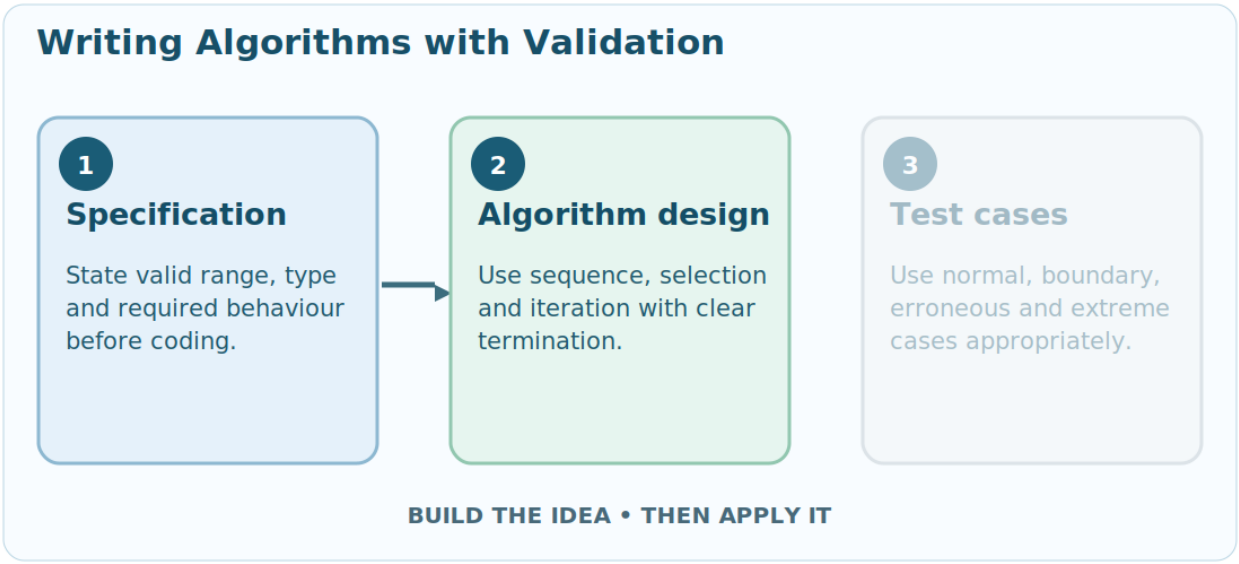
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Writing Algorithms with Validation: Specification then Algorithm design then Test cases

### New Learning

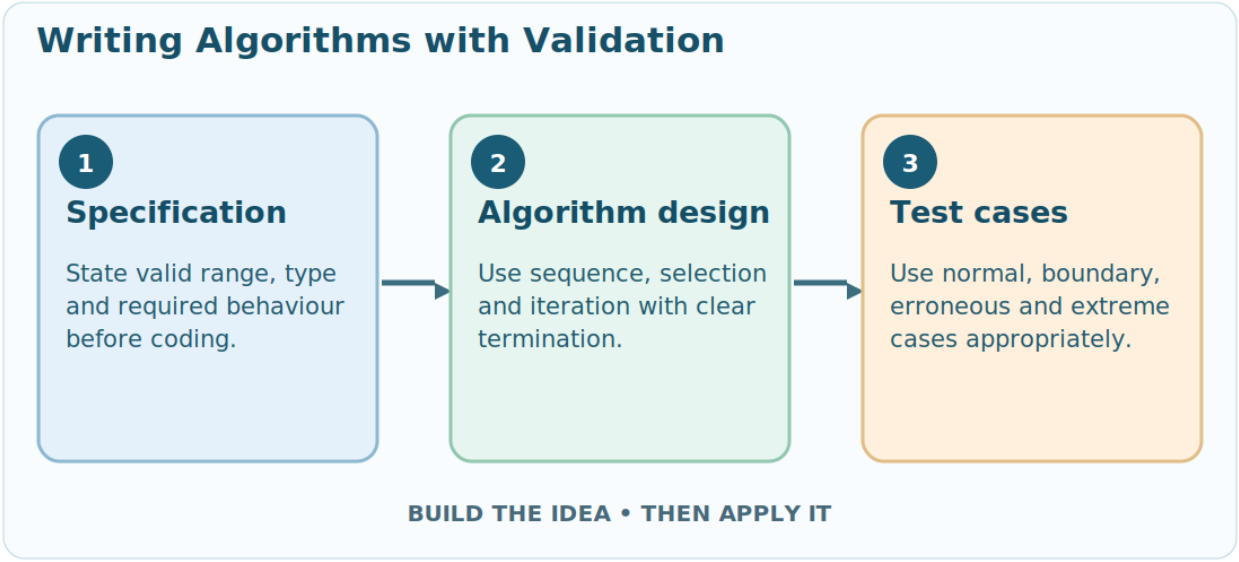



Writing Algorithms with Validation: Specification then Algorithm design then Test cases

# New Learning · original lesson diagrams

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### New Learning




Writing Algorithms with Validation: Specification then Algorithm design then Test cases

I DO — WORKED EXAMPLE

# Follow the worked method

### Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.

### Model answer

```
age = int(input("Age"))

while age < 11 OR age > 19

age = int(input("Age"))

endwhile

print(age)
```

Note: this handles numeric range, not a non-numeric input error without extra handling.

Check: link each claim to the scenario and use precise OCR terminology.

# I Do · original lesson diagrams

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**Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.**

```
age = int(input("Age"))
```



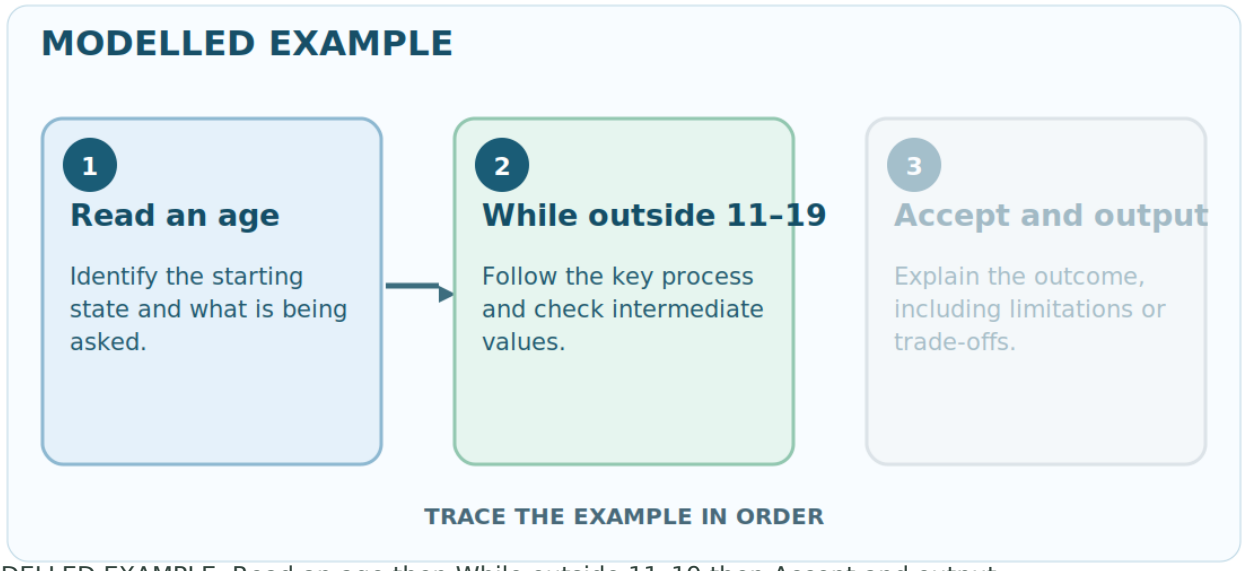

MODELLED EXAMPLE: Read an age then While outside 11–19 then Accept and output

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.**

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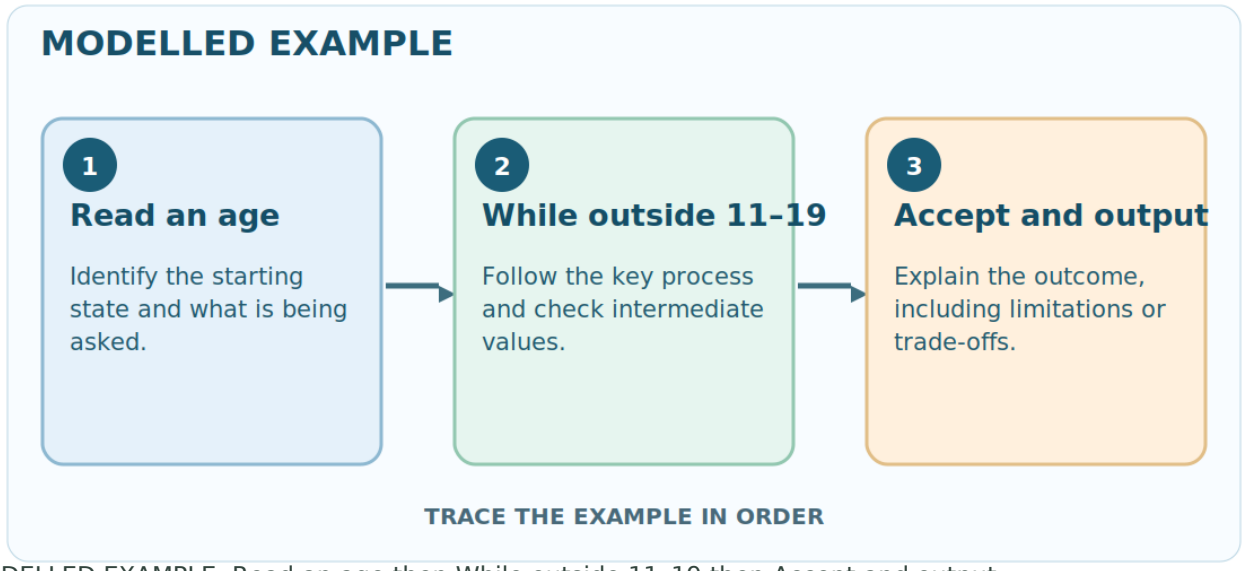

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**Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.**

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age = int(input("Age"))
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MODELLED EXAMPLE: Read an age then While outside 11–19 then Accept and output

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Develop a loop that accepts a score from 0 to 100 inclusive, rejecting out-of-range numbers.**

Identify the inputs, relevant facts and exact output or judgement required.




GUIDED EXAMPLE: Input score then Repeat if <0 OR >100 then Output valid score

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Write an algorithm accepting a positive integer (at least 1) and outputting it. [6]

### Question 2

Explain why 'WHILE age < 11 AND age > 19' fails as range validation. [6]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Extend a validation algorithm to reject non-integer input, describe robust behaviour and supply three tests. [8]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**When rejecting values outside a range, the two invalid inequalities should usually be joined with AND.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

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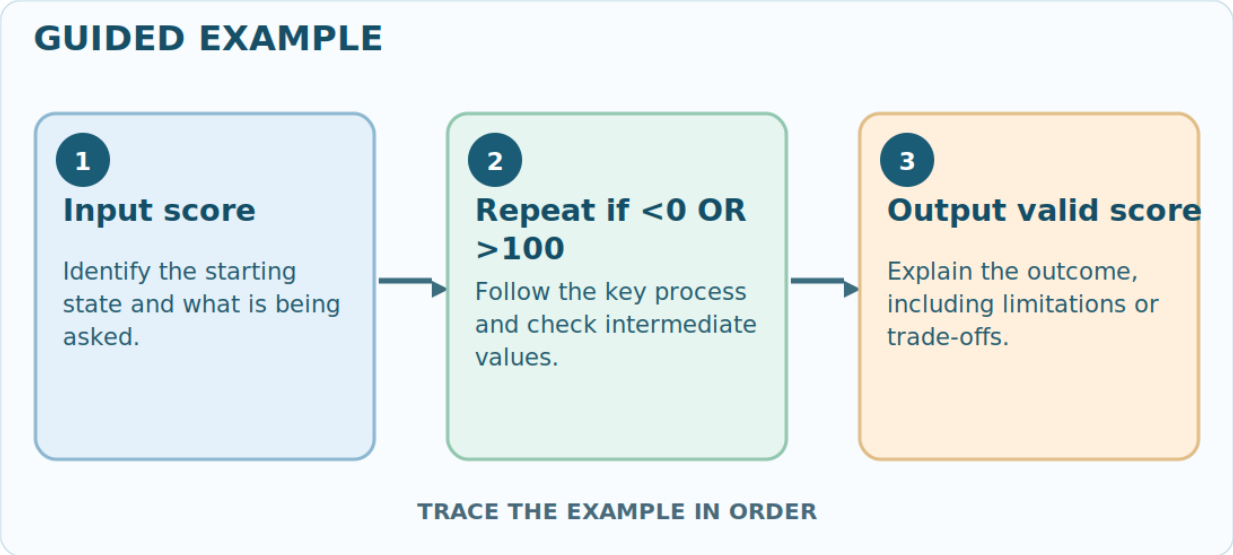

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**Develop a loop that accepts a score from 0 to 100 inclusive, rejecting out-of-range numbers.**

Identify the inputs, relevant facts and exact output or judgement required.




GUIDED EXAMPLE: Input score then Repeat if <0 OR >100 then Output valid score

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

score = int(input("Score"))

while score < 0 OR score > 100

endwhile

print(score)

Explain why OR, not AND, joins the invalid conditions.

IMPROVE: Explain one correction to your own reasoning.